

Study selection

Two reviewers independently screened titles and abstracts, followed by full-text assessment of potentially eligible articles. Disagreements were resolved through discussion and consensus.

Studies were eligible if they: (1) enrolled participants with Tourette syndrome; (2) used a randomized controlled design; (3) compared ecopipam with placebo; and (4) reported extractable efficacy or safety data. Observational studies, case reports, reviews, meta-analyses, editorials, conference abstracts without sufficient outcome data, animal studies, and duplicate publications were excluded. When multiple reports described the same study population, the most complete peer-reviewed publication was retained.

Data extraction

Two investigators independently extracted data using a standardized form. Extracted variables included study characteristics, publication year, sample size, participant demographics, baseline tic severity, treatment duration, intervention details, efficacy outcomes, and adverse events. Any discrepancies were resolved through review of the original reports and consensus discussion.

Outcomes

The primary efficacy outcome was change in Yale Global Tic Severity Scale Total Tic Score (YGTSS-TTS) from baseline to the randomized study endpoint. The YGTSS-TTS is a validated clinician-rated measure assessing tic number, frequency, intensity, complexity, and interference, with higher scores indicating greater tic severity.

Additional efficacy measures, including the Yale Global Tic Severity Scale Global Severity Score (YGTSS-GSS), Clinical Global Impression–Improvement (CGI-I), and Clinical Global Impression–Severity (CGI-S), were extracted and summarized descriptively when available. For safety analyses, headache and insomnia were selected as prespecified outcomes because they were clinically relevant and consistently reported across the included trials.

Statistical analysis

Meta-analyses were performed using R software (R Foundation for Statistical Computing, Vienna, Austria) with the *meta* and *metafor* packages.^{22,23} Continuous outcomes were pooled using the Generic Inverse Variance (GIV) method to calculate pooled mean differences (MDs) with 95% confidence intervals (CIs). Individual studies were weighted according to the inverse of the variance of their effect estimates. When standard errors were not directly reported, they were calculated from the published 95% confidence intervals using established statistical methods. Dichotomous outcomes were pooled as risk ratios (RRs) with 95% CIs.

Random-effects models were applied for all analyses because of anticipated methodological and clinical heterogeneity arising from differences in trial design, treatment duration, and participant characteristics. Hartung–Knapp adjustment was used to provide more reliable confidence interval estimation given the small number of included studies.

Statistical heterogeneity was assessed using Cochran's Q test, the I^2 statistic, and between-study variance (τ^2). Negative MD values favored ecopipam for efficacy outcomes. A prespecified sensitivity analysis excluding the phase 3 randomized-withdrawal trial was performed to evaluate the robustness of the primary efficacy findings and determine whether the pooled estimate was influenced by its enrichment design.

Risk of bias assessment

Risk of bias was independently assessed by two reviewers using the Cochrane Risk of Bias 2 (RoB 2) tool.²⁴ Domains evaluated included bias arising from the randomization process, deviations from intended interventions, missing outcome data, outcome measurement, and selection of reported results. Each study was categorized as having low risk of bias, some concerns, or high risk of bias according to RoB 2 guidance (Figure 2).

Because fewer than ten studies were included, formal assessment of publication bias using funnel plots or statistical tests was not performed. The certainty of evidence was interpreted narratively considering study limitations, consistency of findings, precision of estimates, and the size of the available evidence base.